

Does biological text improve donor-level cell representations?

A controlled, donor-held-out evaluation

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Abstract

Single-cell models increasingly align expression profiles with biological text, but whether the text contributes anything beyond the supervision used to write it is rarely measured. We evaluate this directly with an arm matrix that equalizes cells, donor splits, encoder capacity, update budget and identity-label access: a supervised MLP trained with a direct donor-pathology head (B2), non-text readouts (composition B0; pseudobulk-PCA B1), non-semantic contrastive references (T0 one-hot; T1 random prototypes), and two text-grounded contrastive arms (T2 marker captions; T3 PubMed/RAG captions), plus a permuted-endpoint null (N1) and a permuted caption-map null (N2). Exploratory runs (seeds 0–2) were frozen before a confirmation phase on fresh splits (seeds 3–5 of SEA-AD MTG and ROSMAP, plus a SEA-AD PFC robustness leg), 100 shared donor-permutations per seed, donor-clustered bootstrap intervals, and pre-registered gates. The decisive comparison for a semantic benefit — text arms versus the equally endpoint-blind T1 reference — shows no detectable advantage: confirmatory paired $\Delta\rho$ is +0.041 (T2–T1) and +0.007 (T3–T1) on the primary cohort, both CIs crossing zero. Against direct endpoint supervision the text arms lose: confirmatory SEA-AD MTG ρ was 0.454 (B2) versus 0.261 (T2) and 0.228 (T3), and ROSMAP text arms were significantly *worse* than B2 under paired donor-level tests. In contrast, the source-fitted B2 predictor passed strict cross-cohort transfer in both registered directions (ROSMAP $\leftrightarrow$ SEA-AD MTG, $\rho = 0.40$ – 0.53 over all seeds, gate $\rho \geq 0.35$) and in an MTG $\rightarrow$ PFC same-study robustness check ($\rho \approx 0.77$, donors overlapping). An endpoint-supervision null — encoder retrained on permuted pathology labels, evaluated honestly — retains $\rho \approx 0.3$ – 0.4 , quantifying how much pathology signal the identity/composition channel carries without endpoint supervision; a complementary readout-permuted null on B2’s frozen features centers at zero, verifying the evaluation machinery. Marker-gene recovery against external references failed its gate structurally (only 13–33% of reference genes are evaluable in a 2,000-HVG space). We conclude that under this endpoint and donor scale, contrastive text grounding provides no detectable benefit over a non-semantic reference, and falls short of direct endpoint supervision, while donor-level pathology signal is real and transferable. This is a bounded negative result for the registered comparison, not a claim that biological language cannot help under different designs.

1 Introduction

Single-cell foundation models — scGPT [Cui et al., 2024], Geneformer [Theodoris et al., 2023], scFoundation [Hao et al., 2024] and successors — learn cell representations from expression profiles alone. A parallel line of work argues that grounding those representations in biological language

*Corresponding author. Code and artifacts: <https://github.com/alrobbles/biocellAI>

unlocks generalization that molecular data alone cannot provide: CellWhisperer aligns transcriptomes with AI-curated textual annotations [Schaefer et al., 2025], C2S-Scale renders expression profiles as “cell sentences” inside a general-purpose LLM [Rizvi et al., 2025], and multimodal efforts increasingly treat text as a first-class modality.

The motivating question — *does language grounding improve cell representations?* — is harder to answer than it looks, because “grounding” bundles at least three distinct mechanisms: (i) text can carry *label-discriminative structure* (captions derived from cell-type labels); (ii) text can carry *external biological knowledge* (function, lineage, marker semantics); and (iii) text can act as a *representation bottleneck* that smooths per-cell measurement noise. Published systems typically report end-to-end performance and leave the decomposition implicit — and typically compare text arms against weaker supervision rather than an equally strong non-text baseline.

This paper reports a deliberately controlled decomposition. Earlier exploratory work in this project (protocol v1) appeared to show that caption-grounded embeddings outperform expression-only baselines on donor-level pathology prediction. A methodological audit then identified conflation mechanisms in that record: repeated preprocessing on the same splits, reuse of fold-derived captions at evaluation, transductive PCA, target-label orientation in transfer analyses, and non-equivalent input protocols in external comparisons. Protocol v2 (ADR-011) freezes data contracts, makes every transformation train-only, pits each training signal against an equivalent-supervision control, and evaluates statistically at the donor level — the unit of replication for any donor-held-out claim. Exploratory v2 runs were frozen before a confirmation phase on never-inspected splits.

Our contributions are empirical and methodological:

1. A reproducible donor-held-out benchmark in which cell identity and donor pathology are evaluated under a nine-arm controlled-supervision matrix, verified data lineage, shared permutation nulls and donor-clustered inference (Section 3).
2. A bounded *negative result*: contrastive biological-text alignment (marker captions or retrieved PubMed text) shows no detectable benefit over equally endpoint-blind non-semantic references, and loses to direct endpoint supervision, in either exploratory or confirmatory phases (Section 4.2).
3. A *positive, confirmed result*: a supervised MLP trained on one cohort predicts donor pathology on another cohort with a strictly source-fitted predictor — encoder, normalizations, axis and readout all frozen — in both registered directions and all seeds (Section 4.4).
4. Quantification of a subtle confound: an endpoint-supervision null — encoder retrained on permuted pathology labels, identity head intact, honest readout — retains $\rho \approx 0.3$ – 0.4 , decomposing how much apparent pathology signal the non-endpoint (identity/composition) channels carry (Section 4.3).
5. A structural marker-recovery audit showing the pre-registered retrieval gate was unreachable under the evaluation space, with the attainable ceiling reported rather than the gate redefined post hoc (Section 4.5).

2 Related work

Expression-only single-cell foundation models. scGPT [Cui et al., 2024], Geneformer [Theodoris et al., 2023], scFoundation [Hao et al., 2024], and UCE [Rosen et al., 2023] learn

representations from expression alone; UCE additionally targets cross-species universality via protein-embedding gene tokens. Independent zero-shot evaluations find these models underperform simple baselines [Kedzierska et al., 2025, Atti and Subramaniam, 2025]: Kedzierska et al. [2025] show Geneformer and scGPT lose to HVG selection plus Harmony/scVI on cell-type clustering, and Atti and Subramaniam [2025] report Seurat v5 beating all three on classification. The burden of proof for any added modality is therefore high — motivating the controlled, leakage-aware protocol used here.

Language-grounded cell models. LangCell [Zhao et al., 2024] established contrastive cell-text pretraining on metadata-derived captions; CellWhisperer [Schaefer et al., 2025] scales CLIP-style alignment to $\sim 1\text{M}$ cells with AI-curated descriptions; C2S-Scale [Rizvi et al., 2025] and Cell2Sentence [Levine et al., 2024] serialize expression as “cell sentences” inside general LLMs. None of these decompose which part of the text carries the transferable signal under matched supervision — our central question.

Knowledge-grounded approaches. At the gene level, GenePT [Chen and Zou, 2024] and scGenePT [Istrate et al., 2024] show text-derived gene embeddings are additive but not sufficient alone. Ontology-structured supervision appears in OnClass [Wang et al., 2021], scCello [Yuan et al., 2024], and KCFM [KCFM Team, 2025]; retrieval-based annotation in scRAG [Yu et al., 2025] and scHilda [scHilda Team, 2025]. These ground in *external* knowledge — the direction our results leave open — but target annotation rather than donor-level transfer.

3 Methods

3.1 Data and frozen splits

Three cohorts are used (Table 1). Tabula Sapiens blood [Tabula Sapiens Consortium et al., 2022] (accessed via CZ CELLxGENE Census [CZ CELLxGENE, 2023]) provides a cell-*identity* task (9 donors, 16 cell types). The SEA-AD Alzheimer’s atlas provides donor-*pathology* tasks: a primary middle temporal gyrus cohort (MTG; 84 donors) and a prefrontal-cortex leg (PFC; 80 donors) used only for confirmation and same-study transfer robustness; the endpoint is the continuous cortical pathology score (CPS_Global), a donor-level summary of quantitative neuropathology, not a longitudinal observation. The open ROSMAP multiregion release [Liu et al., 2025] (111 donors) provides an external pathology cohort whose public matrix carries a coarse three-level ordinal pathology label (nonAD < earlyAD < lateAD); we pre-registered this asymmetry against continuous CPS.

Each cohort is reduced to a 2,000-gene highly-variable space selected on *training donors only*, normalized per cell (library size to 10^4 , $\log_1 p$), and split by *whole donors* ($\sim 25\%$ held out). Fold artifacts record source hashes, code hashes, donor lists and the resulting matrix hash; the inference pipeline re-verifies fold SHA-256 values before evaluating gates (integrity gate, Section 3.3). Exploratory folds use seeds 0–2; confirmatory folds use never-inspected seeds 3–5 (MTG, ROSMAP) and 0–2 for the disjoint PFC cell population. All results below report predictions on held-out donors.

3.2 Arm matrix: what is equalized, what is not

All arms see the same cells, the same donor split, the same encoder capacity (a four-layer MLP producing a 64-d cell embedding), the same update budget and the same identity-label access; only the *training signal* differs. Critically, the arms differ in *when and how* the pathology endpoint

cohort	cells	donors	test donors/seed	endpoint	phase
blood (Tabula Sapiens)	85,055	9	2	identity F1	exploratory
SEA-AD MTG	125,909	84	21	CPS ρ	expl. + conf. (s3-5)
ROSMAP	163,646	111	28	path level ρ	expl. + conf. (s3-5)
SEA-AD PFC	119,991	80	20	CPS ρ	confirmatory only

Table 1: Cohorts and endpoints. Test donors differ across seeds, so a donor appears in more than one fold’s held-out set; inference therefore clusters by donor (Section 3.3). SEA-AD PFC donors overlap MTG donors (same study); it is a regional robustness leg, not an independent replication cohort.

is visible: B2 optimizes it directly during representation learning, whereas the contrastive arms (T0–T3) never see it until the frozen donor-level ridge readout.

The matrix therefore supports two distinct comparisons: (a) *semantic benefit at equal endpoint access* — text arms versus the equally endpoint-blind references T0/T1, the clean test of whether caption biology adds anything; and (b) *contrastive text versus direct endpoint supervision* — text arms versus B2, which jointly varies text use and endpoint supervision form. We report both explicitly rather than calling the matrix “equivalent supervision” for the endpoint; a text arm *with* a B2-style endpoint head would isolate a third contrast and is noted as future work.

- **B0** donor composition ($\pm$ covariates) $\rightarrow$ ridge: no encoder, tests how far cell-type proportions alone carry the donor endpoint.
- **B1** pseudobulk expression $\rightarrow$ train-fit PCA $\rightarrow$ ridge: the standard non-encoder baseline.
- **B2** supervised MLP with two heads: cell-type cross-entropy plus a donor-pathology regression head — the non-text reference with equal capacity and label access.
- **T0** contrastive alignment vs. *one-hot* class prototypes: identity structure without semantics.
- **T1** contrastive alignment vs. *fixed random* prototypes: contrastive geometry without biology.
- **T2** contrastive alignment vs. marker-list captions (train-donor Wilcoxon markers, frozen at fold creation).
- **T3** contrastive alignment vs. PubMed/RAG-derived captions — retrieved-literature text rather than engineered prose.
- **N1** B2 with donor $\rightarrow$ pathology labels permuted within training donors: a supervision null that preserves donor structure.
- **N2** the cohort’s stronger text arm (T3 for SEA-AD, T2 for ROSMAP) with the class $\rightarrow$ caption map permuted: a semantics null that preserves text format.

Caption construction. T2 captions are generated per fold: Wilcoxon differential expression on *training donors only* yields the top-8 markers per class, rendered in a fixed template (“*iclassj*: a cell type defined by elevated expression of *jgenesj*”); genes used in captions are logged so downstream evaluations can exclude them. T3 captions come from a pre-registered per-cohort corpus (`type_desc_pubmed_rag_markers_alias.*`): PubMed titles and abstracts retrieved per class over marker-gene queries with HGNC alias expansion, reranked by marker coverage and BM25, and concatenated verbatim (no LLM synthesis in this arm); classes missing from the corpus fall back

to a name-only caption, recorded in the provenance file. Each fold’s caption bank is frozen at fold creation, hashed, and stored with its provenance manifest; N2’s permuted class→caption map uses the same bank.

Donor pathology is read out by a ridge fit on train-donor pooled embeddings and evaluated as Spearman ρ on held-out donors; per-donor predictions are exported for paired inference. Cell identity is read out by a linear probe (and, for contrastive arms, zero-shot nearest-prototype scoring) and evaluated as macro-F1 on held-out cells.

3.3 Statistical protocol and gates

Inference is performed on *paired* per-donor predictions, not on per-seed aggregate metrics:

- **Paired differences.** For each arm comparison, $\Delta\rho$ is computed on donors shared across folds. Confidence intervals come from 2,000 bootstrap replicates that *cluster by donor across seeds* — resampling donors rather than seed×donor cells, because the same donor can appear in several folds’ test sets.
- **Multiplicity.** Primary comparisons (T2/T3 vs. B2 and vs. T1 on the primary cohort, SEA-AD MTG) use empirical permutation p-values; all secondary comparisons carry Benjamini–Hochberg q-values.
- **Permutation nulls (two complementary designs).** 100 shared donor→label permutation maps per seed (300 per cohort) permute labels *among training donors only* and are applied to every arm. (i) *Readout null*: for B0/B1/T0–T3/N2, features are frozen and the donor ridge is fit on the permuted target — a full label null expected to center at zero. (ii) *Endpoint-supervision null*: for B2, the encoder is *retrained* with permuted pathology labels (identity head intact; an N1 replicate) and evaluated with a true-label ridge — this isolates how much endpoint signal the non-endpoint channels carry, and is not expected to reach zero if composition or identity structure carries pathology information. A third check, added in revision, permutes the readout target over the *frozen* B2 encoder’s features to verify the machinery on B2 specifically. Observed ρ is compared against the appropriate null’s 95th percentile.
- **Pre-registered gates** (evaluated once per phase, automatically, from artifacts): integrity (identical fold hashes), improvement ($\Delta\rho \geq 0.05$ over B2 with CI lower bound > 0), semantics (beat T1 *and* exceed the shared null q95), historical-threshold G5b ($\rho \geq 0.5$), transfer (strict $\rho \geq 0.35$), composition-adjusted improvement, and a negative-result-validity check that the full matrix ran with correct provenance.

3.4 Strict cross-cohort transfer

Transfer is *source-fitted*: the donor-level predictor (encoder weights, gene space, train-fit PCA/ridge, label axis and sign) is frozen on the source cohort; the target cohort contributes only cells re-quantified into the source gene space, and no target labels are touched. The gate is $\rho \geq 0.35$ over all target donors. A *calibrated refit* variant (readout refit on target train donors, evaluated on target test donors) is reported separately as an orientation analysis and never counted toward the strict gate. MTG→PFC transfer is additionally flagged as a same-study robustness check: PFC and MTG donors overlap.

arm	exploratory			confirmatory		
	blood	ROSMAP	MTG	ROSMAP	MTG	PFC
B2	0.607	0.949	0.989	0.950	0.993	0.984
T0	0.547	0.899	0.906	0.919	0.913	0.944
T1	0.525	0.878	0.934	0.871	0.937	0.945
T2	0.514	0.918	0.940	0.916	0.965	0.960
T3	0.487	0.677	0.860	0.667	0.857	0.860
N1	—	0.946	0.992	0.950	0.991	0.987
N2	0.498	0.923	0.868	0.898	0.881	0.880

Table 2: Cell-identity linear-probe macro-F1 (mean over seeds). Blood has no pathology endpoint and ran in the exploratory phase only. B0/B1 are donor-level ridge baselines without a cell encoder and are omitted.

3.5 Marker-recovery evaluation

To probe whether biologically meaningful genes drive the caption machinery, we rank genes by train-only Wilcoxon differential expression, *exclude* the genes injected into that fold’s captions (non-circular evaluation), and score the next top-10 against external PanglaoDB marker sets [Franzén et al., 2019] mapped to each cohort’s cell-type vocabulary (P@10 macro over evaluable types). The pre-registered gate was $P@10 \geq 0.6$. We additionally report the *attainable ceiling*: the best achievable P@10 given how many reference genes exist inside the 2,000-HVG evaluation space.

4 Results

4.1 Integrity and identity task

Fold-hash verification passed for all runs in both phases (**integrity:** PASS); every reported number traces to a fold whose source, code and donor lists are manifest-recorded.

All encoder arms learn cell identity on held-out donors (Table 2). On the brain cohorts every encoder arm reaches probe macro-F1 0.86–0.99 (B2 best: 0.95–0.99); on blood (exploratory only) the supervised head reaches 0.61 macro-F1 with contrastive arms at 0.49–0.55 probe / 0.51–0.61 zero-shot. Identity is therefore a solved subproblem at this scale and not the discriminating axis between arms; the decisive comparisons are on donor pathology.

4.2 No detectable benefit of text grounding on donor pathology

Figure 1 and Table 3 report donor-level pathology ρ across arms and phases. Three observations recur in both phases. First, the strongest signals come from *non-text* supervision: B2 and the pseudobulk B1 lead on every pathology cohort. Second, the text arms T2 and T3 sit at or below the non-semantic references T0/T1 on the primary cohort and below them on ROSMAP; they never approach B2. Third, B0 — donor composition alone — carries non-trivial signal (confirmatory $\rho = 0.24$ –0.32), consistent with the composition confound documented in Section 4.3.

Paired donor-level inference (Figure 2) separates the two registered comparisons. *Semantic benefit at equal endpoint access* (text vs. the equally endpoint-blind T1): on the primary cohort, $\Delta\rho(T2-T1) = +0.041$ (95% CI -0.036 to 0.132) and $\Delta\rho(T3-T1) = +0.007$ (-0.125 to 0.162) — statistically indistinguishable from prototypes containing no biology; at 84–111 donors the CIs bound the plausible benefit below roughly ± 0.15 , so modest advantages cannot be excluded but no benefit is detected. *Contrastive text vs. direct endpoint supervision* (text vs. B2): confirmatory

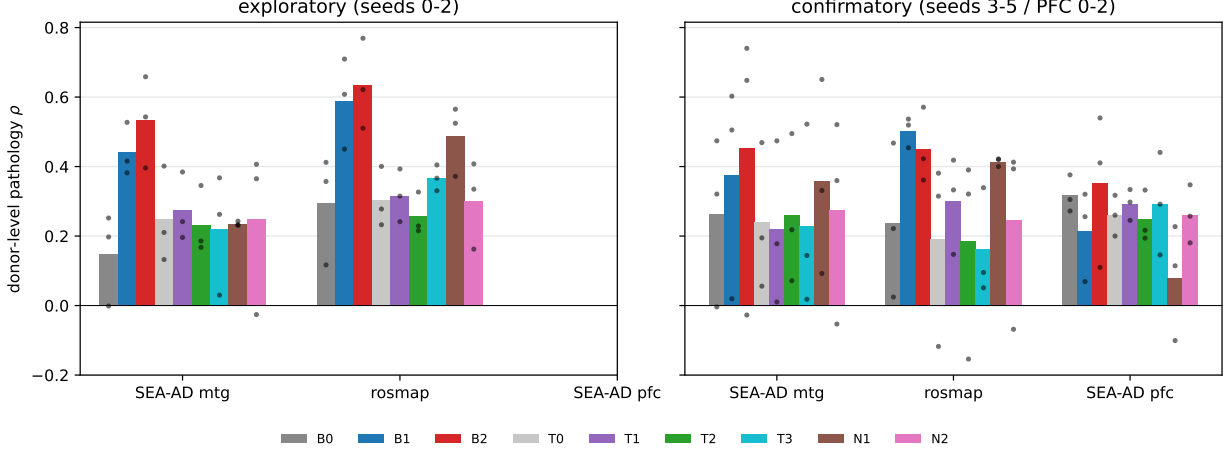


Figure 1: Donor-level pathology ρ by arm and cohort (bars: seed mean; dots: per-seed values). Left: exploratory seeds 0–2. Right: confirmatory seeds 3–5 (PFC: seeds 0–2). Non-text arms (B1, B2) dominate on every cohort; text arms (T2, T3) cluster with the non-semantic references (T0, T1) below them. Blood is absent: it has no pathology endpoint.

	ROSMAP	SEA-AD MTG	SEA-AD PFC
arm			
B0	0.238	0.264	0.318
B1	0.503	0.376	0.215
B2	0.451	0.454	0.353
T0	0.193	0.240	0.259
T1	0.300	0.221	0.292
T2	0.186	0.261	0.248
T3	0.162	0.228	0.293
N1	0.414	0.358	0.080
N2	0.246	0.276	0.262

Table 3: Confirmatory donor-level pathology ρ (mean over seeds 3–5; PFC 0–2). B2: supervised MLP; B1: pseudobulk-PCA; B0: composition; T0/T1: non-semantic contrastive references; T2/T3: marker/PubMed caption grounding; N1: permuted-label retrain; N2: permuted caption map.

$\Delta\rho(T2-B2) = -0.192$ (-0.487 to 0.109) and $\Delta\rho(T3-B2) = -0.226$ (-0.491 to 0.047) on MTG; on ROSMAP the deficit is significant in the exploratory phase ($\Delta\rho \approx -0.27$ to -0.38 , FDR $q < 0.05$) and replicates in sign under confirmation. Because B2 optimizes the endpoint during representation learning and the text arms do not, this second comparison is confounded by supervision form — it bounds what contrastive text achieves relative to the strongest available alternative, not a pure text effect.

4.3 Permutation nulls quantify non-semantic signal

Figure 3 overlays each arm’s observed confirmatory ρ on its shared permutation-null distribution; the two null designs (Section 3.3) answer different questions and must be read separately.

Readout nulls (features frozen, ridge target permuted). For every non-B2 arm these center at zero (grand means $|\mu| \leq 0.01$ across 600+ replicates per arm per cohort), verifying that the evaluation machinery itself produces no signal under label destruction. Observed text-arm ρ (0.19–

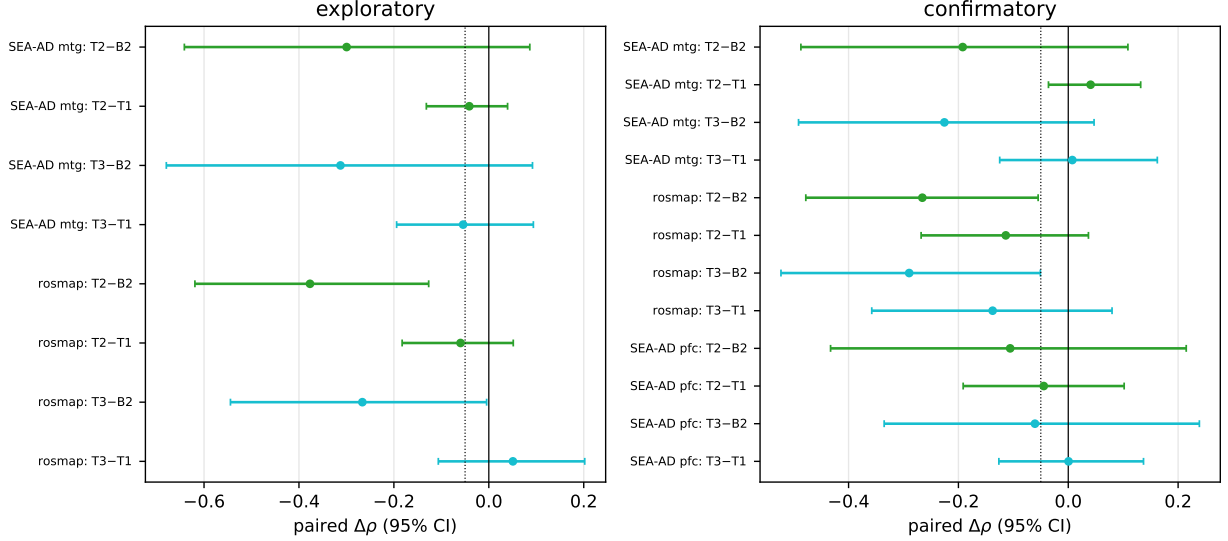


Figure 2: Paired donor-level $\Delta\rho$ (2,000 clustered bootstrap replicates; 95% CIs) for the registered text comparisons. Dotted line at -0.05 marks the improvement-gate boundary in reverse: improvement would require $\geq +0.05$ with $\text{CI} > 0$. Text arms sit at or left of zero against B2 and near zero against the non-semantic T1 reference, in both phases.

0.29) nonetheless lies *below* the q95 of their own readout-permuted nulls (0.32–0.46) — the readout is noisy enough at this donor scale that chance alignments of frozen features with permuted labels can exceed the observed values; the semantics gate fails outright. A verification added in revision permutes the ridge target over the *frozen* B2 encoder’s features on the same shared maps: the resulting null centers at zero (means -0.01 to $+0.02$, q95 0.28–0.41 over 300 permutations per cohort), consistent with all other readout nulls — the machinery is sound on B2’s features too.

Endpoint-supervision null (B2 retrained on permuted endpoint, true-label readout; N1 replicate). This null is deliberately different: it asks how much endpoint signal an encoder *trained without access to the true endpoint* still supports when evaluated honestly. It does not collapse to zero — mean $\rho \approx 0.29$ (ROSMAP), 0.31 (MTG), 0.16 (PFC), q95 0.39–0.59 — because the intact identity head preserves cell-type structure, and donor-level pooling of that structure (composition plus donor-associated expression) correlates with pathology. The in-run N1 arm shows the same effect ($\rho = 0.36$ – 0.41 on ROSMAP/MTG). The correct reading is therefore not “donor identity memorization fakes pathology” but a measured decomposition: roughly two-thirds of B2’s observed ρ is reachable through non-endpoint channels alone, and the endpoint head adds the remainder (≈ 0.1 – 0.15). The cross-cohort strict transfer result (Section 4.4) is the stronger evidence that this remainder captures portable biology rather than within-cohort donor structure.

4.4 Strict source-fitted transfer replicates across cohorts

The positive claim that survives confirmation is B2 strict transfer (Figure 4, Table 4). With predictor, axis and sign entirely frozen on the source cohort, B2 trained on ROSMAP predicts SEA-AD MTG donor CPS at $\rho = 0.46$ – 0.50 (exploratory seeds 0–2) and $\rho = 0.47$ – 0.53 (confirmatory seeds 3–5); the reverse direction reaches $\rho = 0.44$ – 0.50 and 0.40 – 0.50 respectively — every seed above the pre-registered $\rho \geq 0.35$ gate, in both phases. This is a portable, non-cohort-specific pathology signal learned without text.

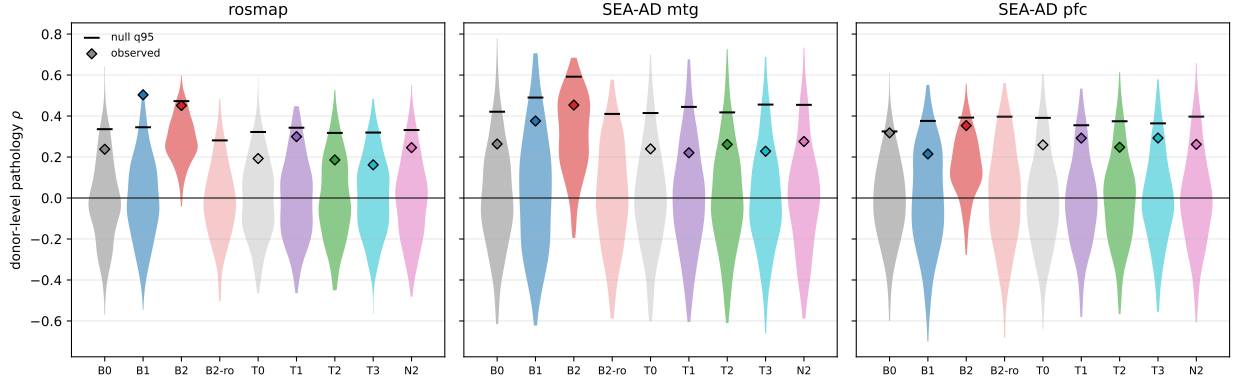


Figure 3: Confirmatory observed ρ (diamonds) against 300 shared donor-permutation nulls per arm (violins; bars mark null q95). Two null designs are shown: for the non-B2 arms the donor ridge is fit on permuted targets over frozen features and centers near zero; for B2 the encoder is *retrained* on permuted endpoint labels and evaluated honestly — its shifted distribution (red, mean ≈ 0.3) measures signal reachable through non-endpoint channels, not a failed null. A separate readout-permuted check on B2’s frozen features centers at zero (text).

Two honest boundaries. First, MTG→PFC transfer is much higher ($\rho = 0.76$ – 0.79 , all confirmatory seeds) but shares donors with the source cohort; we report it as a same-study regional robustness check, not independent replication. Second, the text arms do *not* transfer (strict ρ near or below zero for T0–T3; see generated table), and B1 transfers inconsistently (0.26–0.38, below gate in most seeds). The transferable object is specifically the supervised donor-pathology head, not contrastive geometry.

4.5 Marker recovery: a structural gate failure

Against external PanglaoDB references, fold-derived non-circular marker rankings recover P@10 of only 0.019 (blood), 0.054 (MTG), 0.020 (ROSMAP), 0.024 (PFC) — far below the pre-registered ≥ 0.6 gate. The failure is partly structural: only 13–33% of external reference genes exist inside the 2,000-HVG evaluation space, so the maximum achievable P@10 is 0.16–0.38 — below the gate on every cohort. We report the gate as failed rather than redefining it, with the ceiling documented. Within the attainable range recovery is still weak: brain PanglaoDB references collapse many excitatory subtypes into generic categories, while fold-derived markers are subtype-specific. The caption pipeline’s genes are thus poor external-marker retrievers — consistent with the pathology results, captions here act as a supervision format, not a knowledge channel.

4.6 Gate ledger

Table 5 summarizes the automatic gate evaluation on the confirmatory artifacts. The improvement, semantics and historical gates fail; integrity, strict transfer and negative-result validity pass; composition is not evaluable (see Discussion). Per ADR-011, the complete protocol executed correctly on never-inspected data — the negative finding is a valid experimental outcome, not a pipeline failure.

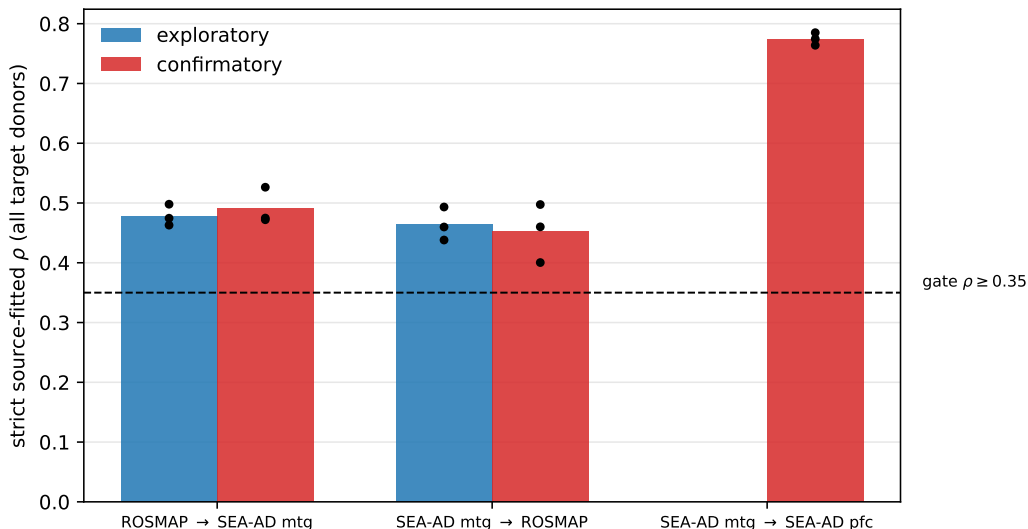


Figure 4: Strict source-fitted transfer of the B2 donor-pathology predictor (all target donors, no target fitting). Bars: seed mean; dots: seeds. Dashed line: pre-registered gate $\rho \geq 0.35$. ROSMAP $\leftrightarrow$ MTG passes in every seed of both phases. MTG $\rightarrow$ PFC (right) is a same-study robustness check — donors overlap — and is reported separately.

4.7 The exploratory record, superseded

For provenance: protocol-v1 exploratory runs had reported apparent text-grounding advantages on the same question (caption-grounded donor embeddings with CPS $\rho \approx 0.60$, cross-cohort transfer, and external-marker retrieval). The audit that motivated ADR-011 showed those estimates conflated the caption channel with (i) repeated preprocessing on the same splits, (ii) evaluation against fold-derived captions whose genes were selected on the same data, (iii) transductive PCA fit before the donor split, and (iv) transfer analyses that used target labels for orientation. Under the corrected pipeline — frozen splits, train-only transforms, equivalent-supervision controls, strictly source-fitted transfer, donor-level inference — the text advantage does not reproduce, while a corrected version of the transfer result does (Section 4.4). We retain the historical numbers only in the repository reports, not as evidence here.

5 Discussion

What the negative result means — and does not mean. The cleanest question the matrix answers is: does aligning a cell encoder to biological captions add anything over equally endpoint-blind, non-semantic contrastive references? The answer, replicated across two independent cohorts and two phases on never-inspected splits, is that we detect none — text arms are statistically indistinguishable from T1, with CIs that bound any benefit to below roughly ± 0.15 in $\Delta\rho$. A second, confounded-by-design comparison shows contrastive text loses to direct endpoint supervision (B2) and to pseudobulk averaging. This is not a claim that language is useless for single-cell modeling, nor that the text arms could not be made stronger: it bounds the benefit of *this* grounding procedure on *this* endpoint at this donor scale. Designs that differ in task (annotation, retrieval), supervision asymmetry (text used to *reduce* label requirements), endpoint access during encoder training (a text arm with a B2-style pathology head, which our matrix deliberately leaves untested), or scale

direction	phase arm	confirmatory	exploratory
ROSMAP $\rightarrow$ SEA-AD MTG	B0	-0.239	-0.239
	B1	0.276	0.384
	B2	0.491	0.478
	N1	0.169	0.158
	N2	-0.055	-0.128
	T0	-0.011	-0.129
	T1	-0.062	-0.267
	T2	-0.027	-0.234
	T3	0.148	-0.194
SEA-AD MTG $\rightarrow$ ROSMAP	B0	0.017	0.017
	B1	0.255	0.317
	B2	0.453	0.464
	N1	0.174	0.112
	N2	0.104	0.074
	T0	0.089	0.083
	T1	0.059	0.111
	T2	0.082	0.080
	T3	0.101	0.086
SEA-AD MTG $\rightarrow$ SEA-AD PFC	B0	0.259	—
	B1	0.320	—
	B2	0.774	—
	N1	0.319	—
	N2	0.291	—
	T0	0.256	—
	T1	0.268	—
	T2	0.255	—
	T3	0.214	—

Table 4: Strict transfer ρ over all target donors (mean over seeds). Only B2 exceeds the gate in every direction and phase. B0 is included for completeness — its composition-only vocabulary transfer is near zero or undefined cross-cohort.

could still show benefits — the contribution here is a protocol capable of detecting them cleanly.

Why supervised and pseudobulk signals win. B2’s advantage plausibly reflects direct optimization of the endpoint: the donor-pathology regression head shapes the pooled representation toward exactly the quantity being read out, whereas contrastive text alignment optimizes a surrogate objective whose information content is bounded by caption quality and register. B1 matching B2 on ROSMAP ($\rho \approx 0.50$ vs. 0.45) further shows that simple expression averaging captures most of what a coarse ordinal endpoint contains — the added value of any encoder, grounded or not, is small when the phenotype is driven by compositional shifts.

The residual-signal confound. That an encoder trained with a *permuted* endpoint still supports $\rho \approx 0.3$ –0.4 under honest evaluation (endpoint-supervision null; N1 replicates it) is the most consequential methodological observation, and it is worth being precise about what it does not prove. Because the identity head remains intact during null training, the retained signal cannot be attributed to donor bookkeeping alone: it measures everything the non-endpoint channels carry — cell-type composition, donor-associated expression, and any residual identity memorization pooled

gate (confirmatory phase)	decision
integrity (fold SHA-256 verified)	PASS
improvement T2 vs. B2 ($\Delta\rho \geq 0.05$, $CI > 0$)	FAIL
improvement T3 vs. B2	FAIL
semantics T2 (vs. T1 and null q95)	FAIL
semantics T3 (vs. T1 and null q95)	FAIL
historical G5b T2/T3 ($\rho \geq 0.5$)	FAIL
strict transfer ROSMAP $\rightarrow$ MTG, MTG $\rightarrow$ ROSMAP, MTG $\rightarrow$ PFC	PASS
composition-adjusted improvement	NOT EVALUABLE
negative result valid (complete protocol)	PASS

Table 5: Confirmatory gate ledger from `experiments/v2_confirmation/inference/gates.json`.

at the donor level. The complementary readout-perm nulls (which destroy labels only at evaluation and do center at zero, including on B2’s own frozen features) rule out an artifact of the readout machinery. Two implications follow. First, donor-level evaluations need both nulls: raw correlations overstate every arm, and retrain-style nulls must be interpreted as channel decompositions, not sanity checks expected to reach zero. Second, our composition gate was registered as **NOT EVALUABLE** in the confirmation phase (covariate/residual block absent from confirmation runs); we report the *presence* of residual signal but claim no composition-adjusted improvement for any arm.

What strict transfer does and does not establish. The source-fitted B2 transfer is the strongest positive evidence in the study: a pathology readout trained on ROSMAP donors ranks MTG donors correctly across every seed of both phases, and vice versa, without touching target labels. Because no target fitting occurs, this is portable signal rather than within-cohort donor-structure memorization — the answer to the confound above. However, both cohorts index Alzheimer’s-related pathology in post-mortem cortex; MTG $\rightarrow$ PFC (additionally same-study, donor-overlapping) and ROSMAP $\leftrightarrow$ MTG together bound the claim to *related-neuropathology* portability, not general biological transfer.

Limitations. (i) Donor counts — not cell counts — bound statistical power; the primary MTG comparisons leave CI widths of ~ 0.5 in $\Delta\rho$, so small true effects are not excluded. (ii) ROSMAP’s public endpoint is a coarse three-level ordinal, asymmetric with continuous CPS; the two pathology tasks are related but not identical. (iii) MTG and PFC share donors; the PFC leg is a robustness check. (iv) Text arms were evaluated at one budget/architecture; a different text encoder or longer training might differ, though the T1 equivalence makes large gains unlikely under this endpoint. (v) Marker-recovery references are coarse for fine brain subtypes, and the evaluation universe (2,000 HVGs) excludes most reference genes. (vi) Blood ran only in the exploratory phase and has no pathology endpoint. (vii) The pre-registered marker-recovery gate was infeasible by construction — a registration-time lesson: gate feasibility should be verified against the evaluation space before registration, and we report the gate as failed rather than silently rescoping it. (viii) The arm matrix equalizes capacity and label access but not endpoint-supervision form; a text arm with a B2-style endpoint head is the natural next experiment and is deliberately out of the frozen protocol’s scope here.

Reproducibility and availability. Code, fold manifests, run manifests, per-donor predictions, permutation-null artifacts, gate JSONs and reports are versioned at <https://github>.

com/alroble/biocellAI (release v0.1.0-alpha, DOI 10.5281/zenodo.23103004), with derived artifacts on Hugging Face (alroble/biocellai-foundations). Controlled-access source data (SEA-AD, ROSMAP raw) are not redistributed; fold contracts record hashes sufficient to reconstruct them from the original repositories. Paper figures and tables are generated from committed artifacts by `scripts/v2_paper_figures.py` — no result is hand-copied.

6 Conclusion

Under frozen data contracts, equal capacity and label access, and donor-level inference with permutation controls, contrastive biological-text alignment shows no detectable benefit over an equally endpoint-blind, non-semantic reference — in exploratory or confirmatory phases, on either of two cohorts — and falls short of direct endpoint supervision. The same protocol confirms that donor-level pathology signal is real and transfers across cohorts when the predictor is strictly source-fitted, and decomposes how much apparent signal the non-endpoint channels carry. The protocol — not a positive text-grounding claim — is the durable result: it converts a literature-level question into an evaluable experiment whose negative outcomes are as reportable as positive ones.

References

- Srijan Atti and Shankar Subramaniam. Fundamental limitations of foundation models in single-cell transcriptomics. *bioRxiv*, 2025. doi: 10.1101/2025.06.26.661767.
- Yiqun T. Chen and James Zou. GenePT: a simple but effective foundation model for genes and cells built from ChatGPT. *bioRxiv*, 2024. doi: 10.1101/2023.10.16.561085.
- Haotian Cui, Chloe Wang, Hassaan Maan, Kuan Pang, Fengming Luo, Nan Duan, and Bo Wang. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods*, 21:1470–1480, 2024. doi: 10.1038/s41592-024-02201-0.
- CZ CELLxGENE. CZ CELLxGENE Discover Census. <https://cellxgene.cziscience.com/census-model>, 2023.
- Oscar Franzén, Li-Ming Gan, and Johan L. M. Björkegren. PanglaoDB: a web server for exploration of mouse and human single-cell RNA sequencing data. *Database*, 2019:baz046, 2019. doi: 10.1093/database/baz046.
- Minsheng Hao, Jing Gong, Xin Zeng, Chiming Liu, Yucheng Guo, Xingyi Cheng, Taifeng Wang, Jianzhu Ma, Xuegong Zhang, and Le Song. Large-scale foundation model on single-cell transcriptomics. *Nature Methods*, 21:1481–1491, 2024. doi: 10.1038/s41592-024-02305-7.
- Ana-Maria Istrate, Donghui Li, Theofanis Karaletsos, and CZ CELLxGENE AI. scGenePT: Is language all you need for modeling single-cell perturbations? *bioRxiv*, 2024. doi: 10.1101/2024.10.23.619972.
- KCFM Team. KCFM: A knowledge-graph-informed foundation model for single-cell annotation, 2025. <https://github.com/nudt-bioinfo/KCFM>.
- Kasia Z. Kedzierska, Lorin Crawford, Ava P. Amini, and Alex X. Lu. Zero-shot evaluation reveals limitations of single-cell foundation models. *Genome Biology*, 26:101, 2025. doi: 10.1186/s13059-025-03574-x.

- Daniel Levine, Syed A. Rizvi, Sacha Lévy, Nazreen Pallikkavaliyaveetil, David Zhang, Xingyu Chen, Sina Ghadermarzi, Ruiming Wu, et al. Cell2Sentence: Teaching large language models the language of biology. In *Proceedings of ICML*, volume 235 of *PMLR*, 2024.
- Zunpeng Liu et al. Single-cell multiregion epigenomic rewiring in Alzheimer’s disease progression and cognitive resilience. *Cell*, 188(18):4980–5002.e29, 2025. doi: 10.1016/j.cell.2025.06.031.
- Syed Asad Rizvi, Daniel Levine, Aakash Patel, Shiyang Zhang, Eric W. Wang, Curtis Jamison Perry, Nicole Mayerli Constante, Sizhuang He, David Zhang, et al. Scaling large language models for next-generation single-cell analysis. *bioRxiv*, 2025. doi: 10.1101/2025.04.14.648850.
- Yanay Rosen, Yusuf Roohani, Ayush Agrawal, Leon Samotorčan, Tabula Sapiens Consortium, Stephen R. Quake, and Jure Leskovec. Universal cell embeddings: A foundation model for cell biology. *bioRxiv*, 2023. doi: 10.1101/2023.11.28.568918.
- Moritz Schaefer, Peter Peneder, Daniel Malzl, Salvo Danilo Lombardo, Mihaela Peycheva, Jake Burton, Anna Hakobyan, Varun Sharma, Thomas Krausgruber, Celine Sin, Jörg Menche, Eleni M. Tomazou, and Christoph Bock. Multimodal learning enables chat-based exploration of single-cell data. *Nature Biotechnology*, 2025. doi: 10.1038/s41587-025-02857-9.
- scHilda Team. scHilda: Hierarchical integration of LLM with KG database for single cell type annotation. *PLoS Computational Biology*, 2025. doi: 10.1371/journal.pcbi.1014291.
- Tabula Sapiens Consortium, Robert C. Jones, Jim Karkanias, Mark A. Krasnow, et al. The Tabula Sapiens: A multiple-organ, single-cell transcriptomic atlas of humans. *Science*, 376(6594): eabl4896, 2022. doi: 10.1126/science.abl4896.
- Christina V. Theodoris, Ling Xiao, Anant Chopra, Mark D. Chaffin, Zeina R. Al Sayed, Matthew C. Hill, Helene Mantineo, Emily M. Brydon, Zexian Zeng, X. Shirley Liu, and Patrick T. Ellinor. Transfer learning enables predictions in network biology. *Nature*, 618:616–624, 2023. doi: 10.1038/s41586-023-06139-9.
- Sheng Wang, Angela Oliveira Pisco, Ariel McGeever, Maria Brbić, Marinka Zitnik, Spyros Darnanis, Jure Leskovec, Jim Karkanias, and Russ B. Altman. Leveraging the cell ontology to classify unseen cell types. *Nature Communications*, 12:5556, 2021. doi: 10.1038/s41467-021-25725-x.
- Zhiyin Yu et al. scRAG: Hybrid retrieval-augmented generation for LLM-based cross-tissue single-cell annotation. In *Findings of ACL*, 2025.
- Xinyu Yuan, Zhihao Zhan, and scCello Team. scCello: Cell-ontology guided transcriptome foundation model. In *NeurIPS*, 2024. Spotlight.
- Suyuan Zhao, Jiahuan Zhang, Zijing Nie, Zizhen Chen, Le Song, et al. LangCell: Language-cell pre-training for cell identity understanding. In *Proceedings of ICML*, volume 235 of *PMLR*, pages 61159–61185, 2024.